Research Proposal 2: From MIL to Hardware, Closing the Gaps in an In-House CubeSat ADCS

SPARRSO Research Fellowship, Notice-02/2026 | Md Tasbeer Ahmed

Summary

I propose to close the open gaps in our ADCS design for NSU-AI-SAT, Bangladesh's entry in the APSCO CubeSat Competition, and to carry the verified algorithms onto our own STM32-based ADCS board, tested step by step up to hardware-in-the-loop.

Item	Detail
Applicant	Md Tasbeer Ahmed, MSc (EEE) student, North South University; ADCS project lead, NSU-AI-SAT
Category	Junior Research Fellow, up to 8 months
SPARRSO priority areas	1 (ADCS), 3 (control theory, sensor fusion), 4 (microcontroller, PCB design)

Background: APSCO and the work already done

NSU-AI-SAT is North South University's 3U CubeSat in the APSCO CubeSat Competition, run by the Asia-Pacific Space Cooperation Organization, of which Bangladesh is a Member State. SPARRSO coordinated the national selection, an APSCO expert panel approved our design, and the team trained at the competition's summer camp in Xi'an on 12–25 August 2026 (Dhaka Tribune). The next APSCO phase is to build, test and validate an engineering model; this proposal delivers its ADCS.

Our ADCS design report and MATLAB/Simulink Model-in-the-Loop (MIL) simulation are complete. TRIAD and a Multiplicative Extended Kalman Filter (MEKF) estimate attitude and gyro bias, and B-dot, quaternion PID and momentum desaturation control six guidance modes. The 3σ pointing budget is 0.096° on the worst axis, 3.1x inside the 0.3° target-acquisition limit.

Gaps found in our design

Our own report leaves seven gaps, each of which blocks a flight-ready ADCS.

Gap	Evidence in our report	How we close it
Undersized magnetorquer	Needs 0.705 A·m²; MTQ200.20 gives 0.20	Select a larger torquer, size our driver for it, re-verify detumbling within 48 h
No attitude fix in eclipse	TRIAD needs the Sun; MEKF then runs on gyro alone	Fuse star-tracker and magnetometer/Earth-sensor measurements in the MEKF
Assumed filter noise	Measurement noise set from an assumed 1° accuracy	Derive it from the sensor noise models
Open dynamic requirements	Drift, jitter and detumble margins still TBC	Add wheel friction and micro-vibration models; 3σ Monte Carlo
Simplified magnetorquer	Modelled as a clamped dipole only	Coil model (resistance, inductance, PWM) matched to our driver
Wheel momentum build-up	Secular momentum margin 0.43x	Schedule desaturation around imaging passes
FDIR and calibration on paper	Described, not implemented	Watchdog, limit and consistency checks, magnetometer calibration in flight code

Objectives

1. Close the seven gaps in MIL and re-verify every ADCS requirement at 3σ.
2. Design, fabricate and bring up an in-house STM32 ADCS board, buying only wheels, magnetorquers and sensors.
3. Generate flight code from the Simulink model and verify it in software-, processor- and hardware-in-the-loop tests.
4. Deliver a tested ADCS for the APSCO engineering model.

Methodology

Each step reuses the MIL test cases, so results stay directly comparable:

1. **MIL:** close the gaps; Monte Carlo at 3σ.
2. **Software-in-the-loop:** C code generated from Simulink runs on a PC against the plant model.
3. **Processor-in-the-loop:** the same code runs on an STM32 development board, checking timing at the 0.1 s estimator step.
4. **Hardware-in-the-loop:** our fabricated board runs closed-loop against the Simulink plant, with sensor data emulated over its real buses and the drivers powering magnetorquer coils.
5. **Bench tests** of purchased wheels and torquers, subject to lab access at SPARRSO or NSU.

Replacing the bought iADCS200 also means redesigning its command and power interfaces around our board.



Figure 1. Verification ladder from simulation to the APSCO engineering model.

Work plan

The gaps are closed first, so the board is built for a verified design.

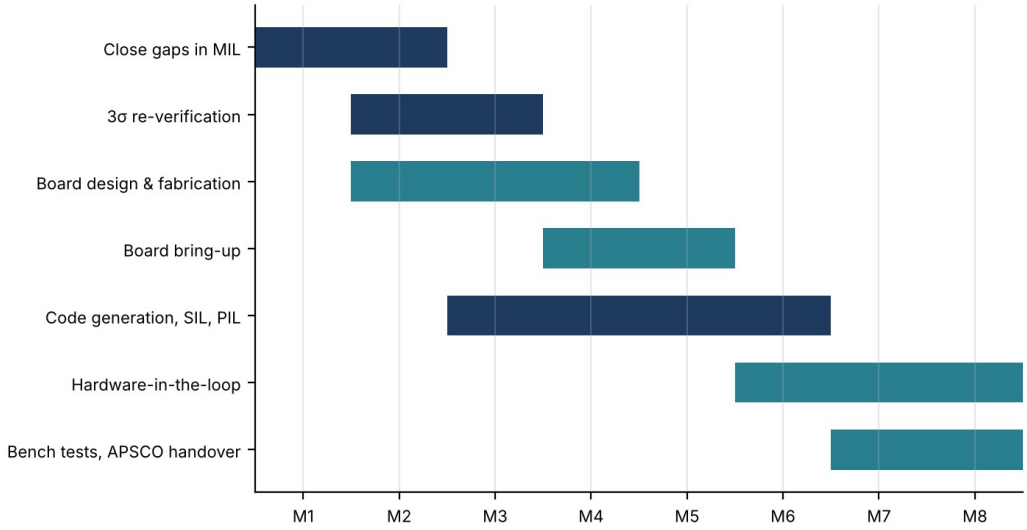


Figure 2. Eight-month work plan (M = month).

Expected outcomes

- A gap-closed ADCS design, every requirement re-verified at 3σ, including pointing through eclipse.
- A working in-house STM32 ADCS board: schematic, layout, bill of materials and bring-up report.
- Flight code generated from the verified model, with software-, processor- and hardware-in-the-loop test records.
- An ADCS ready for APSCO's engineering-model phase.

For Bangladesh, this is a domestically built ADCS that SPARRSO and future APSCO teams can inspect, modify and fly, rather than an imported black box.